

Breno Cordeiro Bispo

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Summary

He received his Ph.D. in Electrical Engineering from the Federal University of Pernambuco (UFPE), Brazil in 2026 (CNPq-funded), with research focused on higher-order signal processing, computational neuroscience, and complex systems. He was a visiting researcher at the Korteweg–de Vries Institute (KdVI) and the Institute for Advanced Study (IAS) at the University of Amsterdam (UvA), The Netherlands, from November 2023 to August 2024 and from July to October 2025, supported by CAPES and the TSP-ARK project. During this period, he actively participated in several academic events, including workshops, seminars, and international conferences.

He received his M.Sc. in Electrical Engineering from UFPE in 2021 (CNPq-funded) and his B.Sc. in Electronic Engineering from UFPE in 2019. He was also a visiting undergraduate student at Hochschule Offenburg, Germany, under the Science Without Borders Program (CAPES-funded), where he studied Elektrotechnik/Informationstechnik from March 2015 to February 2016.

During his undergraduate studies, he served as a teaching assistant for courses in Electrical Circuits, Analog Electronics, and Digital Electronics. His professional experience includes biomedical sensors, embedded systems, Internet of Things (IoT), and FPGA-based hardware design.

His current research interests include (hyper)graph signal processing, topological signal processing, machine learning, neuroscience, and complex systems.

Academic formation

2022 – 2026	Ph.D. in Electrical Engineering – UFPE, Brazil <i>Higher-Order Signal Processing and Its Applications in Brain Network Analysis</i> Advisors: Prof. J. B. Lima (UFPE) & Prof. F. A. N. Santos (UvA) Visiting Researcher – KdVI & IAS, University of Amsterdam (UvA) Supervisor: Prof. F. A. N. Santos (UvA)	<i>granted by CNPq</i> <i>granted by CAPES</i>
Nov. 2023 – Jul. 2024	Visiting Researcher – Univ. of Amsterdam & Universitat Mercatorum (Rome) <i>Project: Topological Signal Processing: Applications in bRain and Knowledge networks (TSP-ARK)</i> Advisors: Prof. S. Sardellitti (Universitat Mercatorum) & Prof. F. A. N. Santos (UvA)	
2019 – 2021	M.Sc. in Electrical Engineering – UFPE, Brazil <i>Vital Sign Monitoring via NFC</i> Advisor: Prof. M. A. B. Rodrigues (UFPE)	<i>granted by CNPq</i>
2015 – 2016	Exchange undergraduate student – Hochschule Offenburg, Germany Course: Elektrotechnik/Informationstechnik	<i>CAPES - Science without borders</i>
2012 – 2019	B.Sc. in Electronic Engineering – UFPE, Brazil <i>Access Control System via RFID/NFC</i> Advisor: Prof. M. A. B. Rodrigues (UFPE)	

Research Interests

Topological Signal Processing • (Hyper)graph Signal Processing • Machine Learning • Neuroscience • Complex Systems
• Internet of Things • Embedded Systems

List of Publications († Most significant publications)

Journal articles under review

- SARDELLITTI, S.; **BISPO, B.C.**; SANTOS, F.A.N.; LIMA, J.B. “Topological Signal Processing Over Cell MultiComplexes Via Cross-Laplacian Operators.” submitted to *IEEE Transactions on Signal Processing*, 2025. <https://arxiv.org/abs/2510.09139>
- † **BISPO, B.C.**; SARDELLITTI, S.; SANTOS, F.A.N.; LIMA, J.B. “Multimodal Higher-Order Brain Networks: A Topological Signal Processing Perspective.” submitted to *IEEE Journal of Biomedical and Health Informatics*, 2026. <https://arxiv.org/abs/2603.29903>
- SILVA NETO, E.F., **BISPO, B.C.**, LIMA, J.B. “Tensor-Based Hypergraph Learning for Schizophrenia and ADHD Classification.” submitted to *IEEE Transactions on Image Processing*, 2026.
- SILVA NETO, E.F., **BISPO, B.C.**, LIMA, J.B. “Tensor-Based Hypergraph Learning for Schizophrenia and ADHD Classification.” submitted to *IEEE Transactions on Image Processing*, 2026.

Journal Articles

- **BISPO, B.C.**, DE OLIVEIRA NETO, J.R., LIMA, J.B., SANTOS, F.A.N. (2026). From pairwise to higher-order brain community detection: A hypergraph signal processing approach on brain functional connectivity analysis. *Computers in Biology and Medicine*. 201, 111409. <https://doi.org/10.1016/j.compbimed.2025.111409>

- † **BISPO, B.C.**; DE OLIVEIRA NETO, J.R.; LIMA, J.B.; SANTOS, F.A.N. “From Pairwise to Higher-Order Clustering: A (Hyper-)Graph Signal Processing Approach on Brain Functional Connectivity Analysis.” *Computers in Biology and Medicine*, 2026. <https://doi.org/10.1016/j.compbiomed.2026.111682>
- **BISPO, B.C.**; DE OLIVEIRA NETO, J.R.; LIMA, J.B. “Hardware Architectures for Computing Eigendecomposition-Based Discrete Fractional Fourier Transforms with Reduced Arithmetic Complexity.” *Circuits, Systems, and Signal Processing*, v.42, 2023. <https://doi.org/10.1007/s00034-023-02493-1>

Conference Papers and Book Chapters

- SARDELLITTI, S., **BISPO, B.C.**, SANTOS, F.A.N., LIMA, J.B. (2025). Cross-Laplacians Based Topological Signal Processing over Cell Multicomplexes. *Proceedings of the 33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, pp. 2647–2651. <https://doi.org/10.23919/EUSIPCO63237.2025.11226604>
- **BISPO, B.C.**, SARDELLITTI, S., SANTOS, F.A.N., LIMA, J.B. (2025). Learning Higher-Order Interactions in Brain Networks via Topological Signal Processing. *Proceedings of the 33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, pp. 2652–2656. <https://doi.org/10.23919/EUSIPCO63237.2025.11226759>
- **BISPO, B.C.**, SANTOS, F.A.N., DE OLIVEIRA NETO, J.R., LIMA, J.B. (2024). Emergence of Higher-Order Functional Brain Connectivity with Hypergraph Signal Processing. *Proceedings of the 32nd European Signal Processing Conference (EUSIPCO)*, Lyon, France, pp. 1332–1336. <https://doi.org/10.23919/EUSIPCO63174.2024.10715376>
- **BISPO, B.C.**, CAVALCANTE, E.L., RODRIGUES, M.A.B. (2024). Access Control System Integrated with RFID and NFC-Enabled Smartphone Technologies. In: *IFMBE Proceedings*, 9th ed., Eds. Springer Nature Switzerland, v.100, p.657–667. https://doi.org/10.1007/978-3-031-49407-9_65
- **BISPO, B.C.**, CUNHA, N.A., SILVA, M.C.B.C., RODRIGUES, M.A.B. (2024). Batteryless NFC Device for Heart Rate and SpO2 Acquisition. In: *IFMBE Proceedings*, 9th ed., Eds. Springer Nature Switzerland, v.100, p.465–475. https://doi.org/10.1007/978-3-031-49407-9_47
- SILVA, A.T.D., **BISPO, B.C.**, ESTEVES, G.R.P., SILVA, M.C.B.C., RODRIGUES, M.A.B. (2022). A Serious Game that Can Aid Physiotherapy Treatment in Children Using Dennis Brown Orthotics. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.445–450. https://doi.org/10.1007/978-3-030-70601-2_69
- **BISPO, B.C.**, CAVALCANTE, E.L., ESTEVES, G.R.P., SILVA, M.C.B.C., ALVES, G.J., RODRIGUES, M.A.B. (2022). Access Control in Hospitals with RFID and BLE Technologies. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.917–924. https://doi.org/10.1007/978-3-030-70601-2_137
- SILVA, A.T.D., **BISPO, B.C.**, CONCEIÇÃO, A.M., SILVA, M.C.B.C., ESTEVES, G.R.P., RODRIGUES, M.A.B. (2022). Adapted Children’s Serious Game Using Dennis Brown Orthotics During the Preparatory Phase for the Gait. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.439–443. https://doi.org/10.1007/978-3-030-70601-2_68
- CUNHA, N.A., **BISPO, B.C.**, CAVALCANTE, E.L., INOCENCIO, A.V.M., ALVES, G.J., RODRIGUES, M.A.B. (2022). Application of MQTT Network for Communication in Healthcare Establishment. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.1465–1470. https://doi.org/10.1007/978-3-030-70601-2_216
- SILVA, M.C.B.C., **BISPO, B.C.**, SILVA, C.M., CUNHA, N.A., SANTOS, E.A.B., RODRIGUES, M.A.B. (2022). Assisted Navigation System for the Visually Impaired. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.1439–1444. https://doi.org/10.1007/978-3-030-70601-2_212
- CUNHA, N.A., **BISPO, B.C.**, FERREIRA, K.R.C., ALVES, G.J., ESTEVES, G.R.P., SANTOS, E.A.B., RODRIGUES, M.A.B. (2022). Communication in Hospital Environment Using Power Line Communications. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.1451–1455. https://doi.org/10.1007/978-3-030-70601-2_214
- SILVA, A.T.D., SILVEIRA, J.V.B.D., ESTEVES, G.R.P., **BISPO, B.C.**, CUNHA, N.A., KUNKEL, M.E., RODRIGUES, M.A.B. (2022). Development and Customization of a Dennis Brown Orthosis Prototype Produced from Anthropometric Measurements by Additive Manufacturing. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.465–470. https://doi.org/10.1007/978-3-030-70601-2_72
- CHAVES FILHO, A.C., INOCENCIO, A.V.M., FERREIRA, K.R.C., CAVALCANTE, E.L., **BISPO, B.C.**, RODRIGUES, C.M.B., LESSA, P.S., RODRIGUES, M.A.B. (2022). Geriatric Physiotherapy: A Telerehabilitation System for Identifying Therapeutic Exercises. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.969–974. https://doi.org/10.1007/978-3-030-70601-2_144
- SILVA, C.M., **BISPO, B.C.**, ESTEVES, G.R.P., CAVALCANTE, E.L., OLIVEIRA, A.L.B., SILVA, M.C.B.C., CUNHA, N.A., RODRIGUES, M.A.B. (2022). Transducer for the Strengthening of the Pelvic Floor Through Electromyographic Biofeedback. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.935–940. https://doi.org/10.1007/978-3-030-70601-2_139
- **BISPO, B.C.**, CUNHA, N.A., CAVALCANTE, E.L., ESTEVES, G.R.P., FERREIRA, K.R.C., CHAVES FILHO, A.C., RODRIGUES, M.A.B. (2022). Wearable System for Postural Adaptation. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.387–393. https://doi.org/10.1007/978-3-030-70601-2_60

Research Projects

2024 – present	Graph Signal Processing: Theoretical Contributions and Application Scenarios <i>PI: Prof. Juliano B. Lima (UFPE)</i> Develops tools that generalise classical signal processing concepts to signals defined over graph-structured domains (sensor networks, brain activity, digital images), in a cooperative network involving Brazilian and international universities. Status: ongoing.
2022 – 2024	Contributions to Signal Processing with Applications to IoT and Digital Image Representation <i>PI: Prof. Juliano B. Lima (UFPE)</i> Advances graph signal processing theory – introducing new tools for signals defined over graphs – targeting applications in IoT sensor networks and digital image representation.
2024 – 2026	TSP-ARK: Topological Signal Processing – Applications in bRain and Knowledge Networks <i>PIs: Prof. Stefania Sardellitti (Universitas Mercatorum) & Prof. Fernando A. N. Santos (UvA)</i> Develops novel topological signal processing methods with applications to brain network analysis.
2020 – 2021	Monitoring of Therapeutic Exercises and Haemodynamic Parameters for Post-COVID-19 Patients <i>PI: Prof. Marco A. B. Rodrigues (UFPE)</i> Developed a wearable device for non-invasive monitoring of vital signs and therapeutic exercises via accelerometry and imaging, targeting remote rehabilitation of post-hospitalisation COVID-19 patients.

Talks & Presentations

- “Network Inference.” *CIMPA Research School – The Mathematics of Complex Systems*, Recife, Brazil, 2026.
- “Application of Topological Signal Processing to Brain Networks.” *Cutting-Edge Perspectives on Brain Networks Workshop*, Universitas Mercatorum, Rome, Italy, 2025.
- “An Overview on Hypergraph Signal Processing and Its Application in Neuroscience.” *KdVI DM Seminar*, University of Amsterdam, 2024.

Professional Experience

Mar.–Dec. 2018	Engineering Intern – EngeBIO NE, Recife · RFID/NFC-based access control system development.
Feb.–Jun. 2021	Teaching Assistant – UFPE · Digital Electronics 1A (internship).
2014 – 2017	Teaching Monitor – UFPE · Electric Circuits (2014); Digital Electronics (2016–17); Analog Electronics (2017).

Technical Skills

Programming	C/C++, Python, MATLAB
Hardware Description	VHDL, Verilog
Dev. Environments	Arduino IDE, Quartus, KiCAD, LTSpice
Microcontrollers	ESP8266, ESP32, ATMEGAs
FPGA Platforms	Altera DE2-115, DE1-SoC
IoT Protocols	MQTT, LoRa, Bluetooth/BLE, WiFi, Ethernet, nRF24L01
OS / General	Windows, Linux

Languages

Portuguese – Native **English** – Fluent (TOEFL ITP B2) **German** – Intermediate (TELC B1)

Awards

2017	SBrT Digital Communications Challenge for Military Applications – Brazilian Telecommunications Society (SBrT).
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